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Book Review:

Filsafat Ilmu dan Logika: Dialektika Perubahan

(Philosophy of Science and Logic: Dialectic of Change)

Author: Lukman Hakim; Number of Pages: 132 Pages; Book Thickness: 15.5 cm; Publisher: Lakeisha Publisher (Member IKAPI No.181 / JTE / 2019); Year of publication: 2020; ISBN: 978-623-6573-47-1.

This book was written based on the question "What and Why is the Philosophy of Science?". Until now, it seems that no one can stop the rapid progress of science. Recent history records science as a form of rational human thought with the results of various scientific disciplines and sub-disciplines. Accompanied by suitable methodological materials and buildings, it is well preserved in libraries or scientific institutions such as universities. The progress of science cannot be separated from every discovery that is useful for the advancement of human life.

The book is perfect for a beginner who wants to study the philosophy of science and wants to know the basics of logic. Very helpful for someone who wants or will study philosophy of science. The content of the discussion that contains the history of philosophical thought to the history of methodology in modern science is beneficial for ordinary people who want to know how to develop science. By studying philosophy of science, we try to look again at a more radical and critical view of the history and development of this book.

The book presents several philosophers in several periods and discusses the essays that have been written. This book contains seven chapters beginning with an introductory discussion of philosophy, several branches of science, the history of the development of philosophy, which has been packaged. In the following chapters, this book discusses the basics of logic, how to present the philosophy of science, to discuss the influence of philosophical thought on modernity.

Reviewing the book entitled "**Filsafat Ilmu dan Logika: Dialektika Perubahan (Philosophy of Science and Logic: Dialectics of Change)**", the reviewer has several reasons he chose this book. Books are by the provisions and themes given by the philosophy of science lecturers. The purpose of reviewing this book is to provide a brief description based on the author's reading experience of the contents in the book. This paper is addressed to someone who will read this book.

The philosophers who have created philosophical ideas have a love for knowledge that is built on the logic of rational thinking. Because the conditions are constantly changing, rational may occupy space and time. The logic that has been used so far is known as deductivism and inductivism is still a human knowledge arena. Scientific logic (*logical scientific*) is an inseparable part of the philosophy of science. Good scientific reasoning equates to minimal logical errors, either an explanation or in conclusion.

The meaning of philosophy consists of two words "*Philo*," which means "love"; "*Shops*" which means "wisdom", which comes from Greek. The difference between a philosopher and someone who only studies philosophical thoughts is that a philosopher is a person who loves wisdom and has authentic (original) thoughts. Meanwhile, people who only study philosophy and philosophical thoughts are called philosophers or observers of philosophy. Because it has a work program or elements such as *systematic*, *radical*, *comprehensive*, *speculative*, and *crisis*, philosophy is called the *mother of science* so that philosophical reflection can give birth to knowledge. Understanding philosophy can be illustrated by the illustration of the philosophical tree (*the three of philosophy*). What are the terms referred to in the philosophy tree? First, ontology, namely the study of philosophy that discusses the *existence* or nature of existence, ontology is called the most ancient philosophy. Second, epistemology is the theory of obtaining knowledge through means, grooves, and strategies of Greek philosophers in justifying knowledge. Third, axiology, namely the study of values, philosophers must be able to defend their opinions so that they can be considered valuable, meaningful, and valuable. Logic comes from Greek, *logos* means orderly mind. Heraclitus is the inventor of the term *logos* for the first time in 534-475 BC. Logic is called Latin; *scientific logic* is the study of the skills to think in an orderly, precise, and rational way. According to experts, there are several benefits of studying logic, among others.

1. Logic is a means of science. Studying science must use the laws of logic because it helps people think correctly and correctly. Therefore, logic is a scientific technique (Sommers, 2013: 3).

2. Increase the ability to express ideas clearly and have weight, develop skills in composing words, and describe in arguing and providing a critical analysis (Sumarnoyono, 1999: 5).
3. Practicing serious thinking, being able to understand the philosophy, and being a good citizen (Kusbandijo, 2010: 12).

There are two kinds of logic, namely syllogism and fallacy. Syllogism is a process of deductive logical reasoning invented by Aristotle. Inference syllogism originates from propositions (which are called premises), which in turn get new propositions (conclusions). The fallacy is a discussion about the causes of humans not based on correct logic. Mistakes that are made intentionally and even unintentionally by humans are often detrimental to themselves and even to others.

The philosophy of science aims to provide a description and reflection of values. This can be done by parsing the flow of history, from the era of Ancient Greece to the *renaissance* (enlightenment century), which finally continued to experience rapid development until now in the heyday of science. The next goal is to provide a philosophical reflection on the development of science. So that what is called science can run according to its capacity to provide a strong foothold to not fall into epistemological violence (*anarchism epistemology*). The benefits that are expected in studying philosophy of science are to provide new insights in looking at the relationship of science and its contribution to human civilization.

Ferdinand de Saussure (1857-1913), as the foundation stone of structuralism through the repertoire of language, used the term structure in linguistic space to designate systems and authority (Bertens, 1996: 180). Saussure shared the distinction of the term which later, in linguistic history, played an important role in structuralism. This difference is a significant and sign (signifies). In language terminology, the significance is a meaningful sound which is a material aspect of language. Meanwhile, Significance is a mental description of the language that is a substance and a significant driving force.

The last discussion reaches a social construction that can be understood in the 21st century which is still worthy of being called the modern era, except that it has a new phase marked by significance radical process between globalization and localism, in the term of modern sociology, namely *glocal*. The Indonesian democratic system had experienced its heyday during the Parliamentary Democracy in 1950-1959. And Indonesia has also experienced a democratic period like a bureaucratic state. Characteristics, a group of elites fully control the political power of the state. Meanwhile, the community is only involved in the implementation of power.

The review writer here only provides a summary of the discussion in this book. Maybe those of you who want to read this book know a little about what is discussed in this book. The book discusses many of the ideas of philosophers and their influence in the modern era, if you want to find out please feel free to read this book in full. An easy-to-understand discussion can provide insight into why the philosophy of science and logic of thinking is very beneficial for human life.

In the book there are still words or sentences that are confusing or there seems to be a typo, these words are "As a food research procedure using a method" and "Therefore, moral is very necessary with subjectivity". According to the author, the rest has been very helpful in providing knowledge about the philosophy of science, the initiators of philosophy, and logic that can be applied in everyday life.

In conclusion, this book is very helpful for someone who wants or will study philosophy of science, arguably a beginner. The content of the discussion which contains the history of philosophical thought to the history of methodology in modern science is very useful for ordinary people who want to know how the development of science. This book can also be used as a reference in modern times like today.

As a refinement in this work, the writer should explain the discussion of syllogism. As there are formulas that are not known by ordinary people and there are patterns that must be thought deeply to understand them. From these suggestions, a more detailed explanation should be provided in the next print. For the whole book, this is good, only in that part that a little confuse the reader so that it passes the discussion.

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Lukman Hakim, born in Madiun on January 3, 1988. He graduated from SMA N 1 Geger Madiun in 2006 and continued his studies at Gajah Mada University at the Faculty of Philosophy undergraduate level. In 2016 Lukman Hakim graduated with a Master of Philosophy degree using the BPP-DN (Domestic Postgraduate Education Scholarship) by the Directorate General of Higher Education (Dikti). Currently teaching at the University of 17 August 1945 Surabaya by teaching courses in the fields of Philosophy, *Social Humanities*, Pancasila Education, and Citizenship Education.

